

SQL Speedinator Report

■ ■ Target Server: TEST-SERVER

Report Generated	2025-12-26 20:49:25
Analysis Duration	0.0 seconds
Databases Analyzed	N/A
Report Version	2.0 - AI Enhanced

Performance Analysis Report - Powered by AI

■ Executive Summary

■ ■ Server Information

Data	{'server_name': 'TEST-SERVER', 'sql_version': 'Micro
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■ ■ Wait Statistics Analysis

No wait statistics data available.

■ Index Analysis

Issues Found:

Index maintenance analysis found 0 indexes requiring rebuild, 0 needing reorganization, 1 unused indexes, and 1 overlapping/duplicate indexes that could be optimized.

■ Overlapping & Duplicate Indexes

Schema.Table	Index 1	Index 2	Type	Key Columns
.Orders	Unknown	Unknown	■ Unknown	Unknown

■ Missing Index Analysis

Issues Found:

Identified 4 missing index opportunities. 2 have high performance impact and should be prioritized.

Database.Table	Equality Columns	Impact	Seeks	Priority
TestDB.Orders	customer_id	85.5	10000	HIGH
TestDB.Products	category_id	72.3	8000	HIGH
TestDB.Customers	city	45.2	3000	MEDIUM
TestDB.Archive	status	15.1	500	LOW

■ AI-Powered Performance Analysis

Executive Summary

This SQL Server instance shows classic signs of parallelism contention and I/O pressure. The combination of high CXPACKET and PAGEIOLATCH waits, along with significant index fragmentation, suggests immediate attention to MAXDOP configuration and index maintenance strategy. Consider implementing automated maintenance and monitoring query execution plans for optimization opportunities.

■ AI-Identified Performance Bottlenecks

Priority	Bottleneck Issue	Impact	AI Recommendation
#1 (CRITICAL)	High CXPACKET waits (25.5%) indicate excessive parallelism causing thread contention	HIGH	Review MAXDOP setting and cost threshold for parallelism. Consider setting MAXDOP to number of cores per NUMA node.

#2 (CRITICAL)	PAGEIOLATCH_SH waits (15.2%) suggest I/O subsystem bottleneck	HIGH	Investigate disk performance, consider adding more memory, and optimize slow queries causing excessive I/O.
#3 (MODERATE)	High index fragmentation (45.2%) on IX_Products_Category affecting query performance	MEDIUM	Implement regular index maintenance schedule and rebuild highly fragmented indexes during maintenance windows.

AI Analysis powered by Azure OpenAI • Results should be validated by database professionals